Reduced efficiency in applying Binding Principle B during L2 English pronoun resolution

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Introduction A key question for L2 processing research is whether non-target-like processing behaviors is due to misrepresentation of target structures or inefficient use of linguistic cues (e.g., [1], [2], [3]). The present study builds on prior L1 English studies ([4], [5]) and explores whether Binding Principle B ([6]) similarly constrains the retrieval of structurally accessible antecedents in L1 and L2 English speakers (Chinese as L1). Additionally, we extend our research to include L1 Chinese as a baseline to assess the potential role of L1 transfer during L2 processing.

Previewing our results, we found similar processing patterns in L1 Chinese and L1 English as initially L1 speakers only considered structurally accessible **target** antecedents. By contrast, L2 English speakers considered both structurally accessible targets and inaccessible **distractors**. Furthermore, as L2 proficiency increased, interference from the structurally inaccessible distractor also increased at later processing stages when the distractor was the only semantically plausible referent. However, L2 speakers' native-like acceptability judgments suggest that their non-native processing patterns are unlikely due to deficits in syntactic representation.

Methods We crossed the factors Target Congruency (Target Match/Mismatch) and Distractor Congruency (Distractor Match/Mismatch) for **Exp.1** (L1 Chinese, N = 72), **Exp.2** (L1 English, N = 66), and **Exp.3** (L2 English N = 66). Gender was chosen to establish congruency between the pronoun and the referent. See (1) for an example. The target refers to the structurally accessible referent, and the distractor refers to the structurally inaccessible referent.

Participants finished a self-paced reading task (**SPR**) and indicated their acceptability of the sentences on a 1-7 Likert scale (**AJ**). L2 participants additionally completed two cloze tests. AJ ratings were used to examine syntactic representations; reading times (RTs) were analyzed to assess real-time antecedent retrieval. We used the **gender mismatch effect** – longer RTs in case of gender mismatch between the pronoun and a referent – as a diagnostic tool to test whether participants attempt to establish a dependency/coreference. The scores of cloze tests were taken as measures of L2 proficiency. Twenty-four target items plus 36 fillers were created. Half of the fillers were grammatical and half ungrammatical (attention checkers).

Results Mixed effects linear regressions were used to analyze AJ ratings (**Table 1**) and RTs (**Fig.1-3**). The analysis of the AJ ratings indicates that L2 English speakers were very similar to L1 English speakers in showing a main effect of Target Congruency only ($p < 0.01$), suggesting native-like syntactic representations in L2 English. For the analysis of RTs, in the L1 Chinese baseline **Exp.1**, we discovered a main effect of Target Congruency ($p < 0.001$) as gender mismatch between the target and the pronoun led to slower RTs. At the 1st spillover region, we also identified a main effect of Distractor Congruency ($p < 0.05$), in addition to a main effect of Target Congruency ($p < 0.05$). The distractor effect was due to longer RTs when the distractor semantically matches the pronoun (i.e., a gender match effect). These results show that native speakers initially only considered targets. In the L1 English baseline **Exp.2**, we similarly found gender mismatch effects at the pronoun and the 1st spillover region (ps < 0.05), as well as a gender match effect at the 1st spillover region ($p < 0.05$). However, in the L2 English **Exp.3**, we found main effects of both Target Congruency and Distractor Congruency at the pronoun region (ps < 0.05), which suggests that L2 speakers simultaneously considered structurally accessible and inaccessible referents at an earlier stage, compared to L1 speakers. Together with the AJ data, L2 speakers seemed less efficient in retrieving target antecedents. Proficiency analyses further revealed that more proficient L2 speakers were increasingly affected by distractor match at the spillover region when the distractor was the only semantically plausible antecedent.

Conclusion We highlight two main findings. First, Binding Principle B similarly constrains real-time pronoun resolution in L1 Chinese and L1 English at the early stages. Second, despite native-like L2 representation, L2 English speakers show reduced efficiency in using structural cues to identify accessible antecedents.

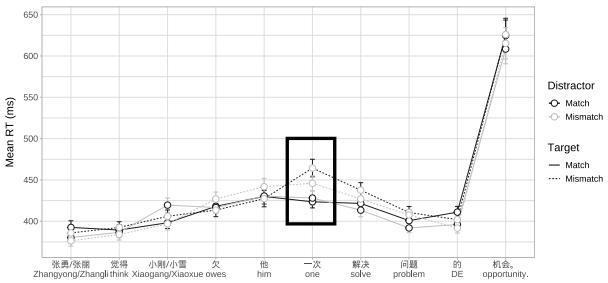
(1) An example item (Chinese and English sentences largely matched in meaning; we only show an English sentence here).

Target Match conditions: **John**$_{\text{target-match}}$ believed that **Peter**$_{\text{distractor-match}}$/**Mary**$_{\text{distractor-mismatch}}$ owed **him** another chance to solve the problem.

Target Mismatch conditions: **Jane**$_{\text{target-mismatch}}$ believed that **Peter**$_{\text{distractor-match}}$/**Mary**$_{\text{distractor-mismatch}}$ owed **him** another chance to solve the problem.

Table 1. Mean acceptability by condition and participant group (SE in parentheses).

	Target Match		Target Mismatch	
	DistractorMatch	DistractorMismatch	DistractorMatch	DistractorMismatch
L1 Chinese	5.88 (0.06)	5.98 (0.06)	5.77 (0.06)	5.89 (0.06)
L1 English	5.77 (0.08)	5.80 (0.08)	**5.16*** (0.09)	**5.10*** (0.09)
L2 English	5.29 (0.09)	5.28 (0.09)	**4.47*** (0.09)	**4.39*** (0.09)



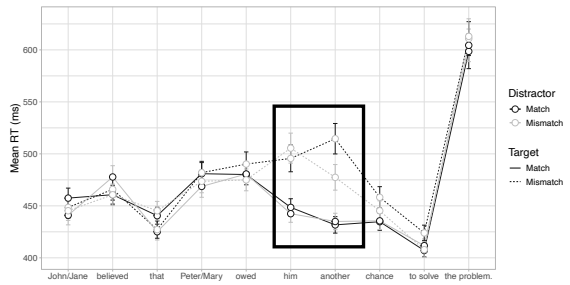
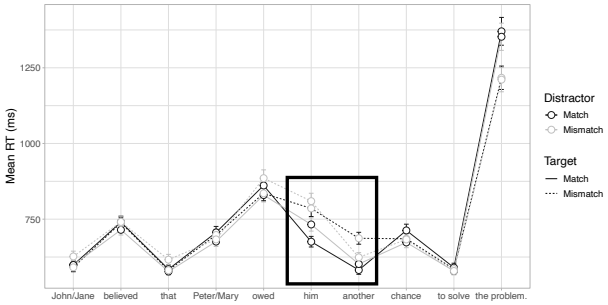


Fig.1 (left): RTs in Exp.1 (L1 Chinese); **Fig.2** (right) : RTs in Exp.2 (L1 English).



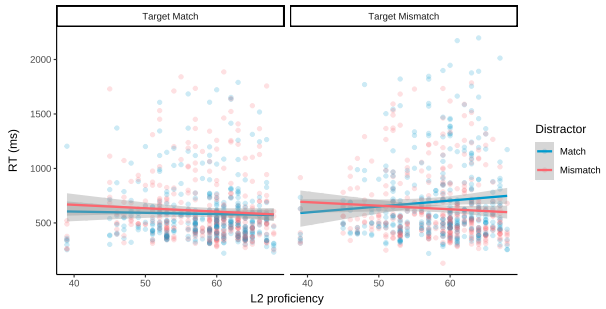


Fig.3 (left): RTs in Exp.3 (L2 English); **Fig.4** (right): Correlation between L2 proficiency and distractor interference effect.

References

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